

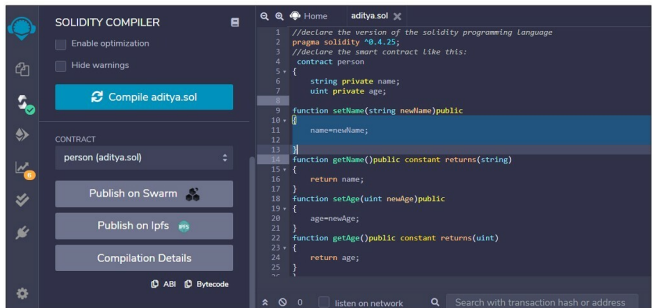


Figure 1: Compiling a smart contract with the Remix IDE

code snippets with Eth Fiddle.

## Writing Solidity programs for blockchain applications

To start writing a program in Solidity, we first need to define the pragma directive:

```
pragma solidity ^0.4.25;
```

The above statement means the contract compiles with a compiler version beginning with 0.4.25. Next, to define a contract, the contract keyword can be used:

```
contract person {
}
```

The complete Solidity program for setting name and age variables is given below:

```
$ //declare the version of the solidity
programming language
pragma solidity ^0.4.25;
//declare the smart contract like this:
contract person
{
    string private name;
    uint private age;
    function setName(string newName) public
    {
```

```
        name=newName;
    }
    function getName()public constant
    returns(string)
    {
        return name; }
    function setAge(uint newAge)public
    {
        age=newAge; }
    function getAge()public constant
    returns(uint)
    {
        return age; }
}
```

## Steps to compile the Solidity program

To compile a smart contract, the easiest online available IDE solidity compiler

is Remix IDE, which is available at <https://remix.ethereum.org>. When you first load the Remix IDE, you'll see a default page, as shown in Figure 1. On the left side of the window, you'll see a tab named *Compile*. Click on it.

Next, deploy the smart contract on a local blockchain. In the toolbox panel, go to the 'Run' tab. This is used to deploy and interact with smart contracts (click it to enlarge the image). For the environment, make sure to choose 'JavaScript VM'. This means that Remix will deploy your smart contract to a local Ethereum blockchain. This blockchain is not connected to the real Ethereum blockchain and does not use real Ether. In this way the smart contract is successfully compiled using the Remix IDE. 

## References

- [1] <https://remix.ethereum.org/>
- [2] <https://studio.ethereum.org/>
- [3] <https://solidity.readthedocs.io/en/v0.7.1/>

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